Nanyang Technological University
School of Electrical & Electronic Engineering
IE2106 Engineering Mathematics I

Tutorial 1 Probability & Statistics

1. (a) We want to choose one number at random from the digits 1, 2, 3. What is the probability that we get an even number? An odd number? Write down the sum of the two probabilities, what does that tell you?

   (b) Three numbers are to be chosen at random from the digits 1, 2, 3 *with replacement*. That is, the three digits are drawn, one following the other, with replacement. What is the probability that

   (i) none of the three digits is even?

   (ii) exactly one is even?

   (iii) exactly two are even?

   (iv) all three are even?

   Identify the rules you are using in each part of the question.

Answers

1.(a) $\frac{1}{3}$, $\frac{2}{3}$ (b) $\frac{8}{27}$, $\frac{4}{9}$, $\frac{2}{9}$, $\frac{1}{27}$